

Limiting Lab Revised CLASS SET

Safety:

- Solid powdered sodium carbonate is an eye and respiratory tract irritant. It is also slightly toxic by ingestion. Avoid contact and wash hands after handling it. If exposed, remove victim to fresh air. If in eyes, rinse with cold water for up to 15 minutes.
- Sodium carbonate is a strong base when dissolved in water and can cause irritation of the skin. Avoid contact and wash hands after handling it. If in eyes, rinse with cold water as soon as possible.
- Calcium chloride solution is slightly toxic by ingestion. Call a physician immediately if ingested.

FOR THIS LAB: Be sure to set up your full data table and complete ALL questions in your lab book analysis section.

- a) Gather together the following: a 25 mL graduated cylinder, two 100 mL beakers, a stir stick, funnel.
- b) Wash all equipment with a test-tube brush & tap water. Dry equipment thoroughly with paper towel.
- c) Set up your retort stand for a filtration (i.e. you should have a ring clamp and funnel in place).
- d) Using tape & a pen, label the 100 mL beakers "sodium carbonate" and "calcium chloride dihydrate".
- e) Weigh APPROX 1.00 g sodium carbonate & 2.00 g calcium chloride dihydrate into separate 100 mL beakers. RECORD THE EXACT MASS USED
- f) Add approximately 25 mL of distilled water to each 100 mL beaker. Gently swirl the contents of the beakers to dissolve the solids. All solid should be dissolved before you proceed (this will take 1 – 2 minutes).
- g) Pour the CALCIUM CHLORIDE solution INTO the sodium carbonate solution, directing the stream with the glass stir rod. You will see calcium carbonate form. SWIRL this mixture gently for 10 – 20 seconds.
- h) Get one or two pieces of filter paper (teacher will tell you). **Weigh and record the mass.**
- i) With the pieces of filter paper together, fold them to make a cone. Insert into funnel and wet with distilled water to stick it to the sides.
Place the empty beaker under the funnel. Swirl the mixture & pour it into the funnel. Add 3-5 mL of distilled H₂O to the beaker that held the mixture. Using a spatula, thoroughly scrape the sides and bottom of the beaker to dislodge all remaining residue. Pour this liquid into the funnel. Repeat this rinsing procedure (3-5 mL distilled H₂O and scraping) twice more to ensure that you have all of the calcium carbonate residue.
- j) If the filtrate appears cloudy, pour the filtrate back into the funnel to filter the liquid for a second time. If the filtrate is clear, carefully remove the filter paper (with solid) and lay it on a piece of paper towel. Put your name on the paper towel and leave it on the tray at the front of the room. Put your name on the beaker containing the filtrate and leave it at the front of the room.

Summary for Precipitation Reaction

Answer the questions below before moving on.

1. Write the **balanced** chemical equation, indicating states, for this reaction of calcium chloride dihydrate with sodium carbonate to form calcium carbonate and sodium chloride, the dihydrate is represented on the product side as 2 solitary water molecules. The calcium carbonate (chalk) is the precipitate. Check that your equation is correct.
2. Predict the mass of **calcium carbonate** that will form in your experiment. Which chemical is the limiting reagent in this lab? (show calculations for all questions)
3. How many grams of NaCl should be produced from today's lab? CaCO_3 ?
4. Say you used one gram more of the **sodium carbonate** than you did. How would that affect the amount of **calcium carbonate** you *calculate* will be formed? How would it affect the amount you actually collect?
5. If you used water that already had some **sodium carbonate** dissolved in it how would that affect the mass of **calcium carbonate** that you actually collect? How would that compare to the amount of **calcium carbonate** you *calculated* would result from the reaction? Explain.
6. If you dissolve the **sodium carbonate** in water that already had some **calcium chloride** in it how would that affect the mass of **calcium carbonate** that you actually collect? How would that compare to the amount of **calcium carbonate** you *calculated* would result from the reaction? Explain.
7. **Check-in** _____ When you have completed this objective check in with your teacher. This is a required part of the lab and your teacher's initials are required before you can move on to the next objective.

Day 2 Determine your yield

I) Keep your filter paper (with product) and filtrate until you have answered all of the questions below. Weigh the filter paper containing the dry CaCO_3 . Be careful not to spill any solid. Calculate the mass of CaCO_3 produced:

1. Referring to your calculations above, which reactant is there an excess of?
 - a. Where can it be found?
 - b. Design and carry out an additional step to prove that you correctly identified which chemical is in excess. Describe your procedure and the result.
2. Is the mass you measured close to the expected mass you calculated based on stoichiometry? Calculate a percent difference between the two.
3. Would the mass of calcium carbonate that you measure be made larger or smaller if it were not completely dry? Explain.
4. Would the mass of calcium carbonate that you measured be made larger or smaller if you washed the precipitate in the filter paper before completely drying it? Explain.
5. Explain any differences between your measurement and the calculation of the mass of calcium carbonate. In particular, comment on why the percent yield was (likely) not 100%

WHEN DONE: Throw the filter paper in the garbage, empty the filtrate into the waste beaker, clean and return all equipment

